MIL-T-11292C
9 January 1979
SUPERSEDING
MIL-T-11292B
18 June 1958

MILITARY SPECIFICATION

TURNOUT, RAILWAY (UNASSEMBLED, COMPLETE WITH LOW STAND THROW AND RIGID BOLTED FROG)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers unassembled turnouts for use with 90- thru 132-pound (41- thru 65-kg) rails to divert railway rolling stock from one track to another.

1.2 Classification. Turnouts shall be of the following sizes for use with a weight of rail as specified (see 6.2):

Number 6.
Number 8.
Number 10.
Number 12.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 2250

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-R-3518	- Rails, Tee, Railway.
MIL-C-3774	- Crates, Wood; Open, 12,000 and 16,000 Pound Capacity.
MIL-B-3964	- Bolts and Nuts, Track.
MIL-T-12270	- Tie-Plates, Railway.
MIL-J-12368	- Joint Bars, Rail.

STANDARDS

FEDERAL

FED. TEST METHOD STD. No. 101	- Preservation, Packaging, and Packing Materials, Test Procedures.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-615	- Turnouts, Railway, with Bolted Rigid Frogs.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN RAILWAY ENGINEERING ASSOCIATION (AREA)

AREA Manual.

(Application for copies should be addressed to the American Railway Engineering Association, 59 East Van Buren Street, Chicago, IL 60605.)

3. REQUIREMENTS

3.1 Description. The turnouts shall consist of a switch assembly and a rigid bolted frog.

3.2 Material. Material shall be as specified herein. Material not definitely specified shall be in accordance with AREA specifications (see 6.5).

3.3 Components. The turnout components shall be fabricated in accordance with MIL-STD-615 and as specified herein, except that ties, spikes, and rails other than switch rails shall not be included. When specified (see 6.2 and 6.3), closure rails shall be included.

3.3.1 Switch assembly. The switch assembly shall consist of a low switch stand, connecting rods, switch rails, and all other parts necessary for a complete assembly. Unless otherwise specified (see 6.2), the switch stand shall be ground throw, automatic safety type of standard commercial design having adjustable throw and a reflex switch lamp not illuminated and without day signal target.

3.3.1.1 Rails. Rails shall conform to MIL-R-3518, weight as specified (see 6.2).

3.3.1.2 Bolts and nuts. Track bolts and nuts shall conform to MIL-B-3964, size as specified (see 6.2).

3.3.1.3 Joint bars. Joint bars shall conform to MIL-J-12368, size as specified (see 6.2).

3.3.1.4 Tie plates. Tie plates shall conform to MIL-T-12270, size as specified (see 6.2).

3.3.1.5 Guard rails. Unless otherwise specified (see 6.2), guard rails conforming to AREA specifications shall be provided.

3.3.2 Frog. The frog shall be of the rigid, bolted type and shall conform to AREA specifications for frogs.

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3.3.3 Closure rails. Closure rails shall conform to AREA specifications for closure rails.

3.4 Identification marking. Each turnout shall be identified in accordance with MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that materials used are manufactured and examined in accordance with referenced standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.4).

4.3 Quality conformance inspection.

4.3.1 Examination. Each turnout shall be examined for the following defects:

101. Material not as specified.
102. Components not as specified.
103. Improper identification marking.
104. Missing or improper parts and components.

Units having one or more defects shall be rejected.

4.4 Inspection of packaging.

4.4.1 Preproduction pack inspection.

4.4.1.1 Examination. The preproduction pack shall be examined for the defects specified in 4.4.2.3. Presence of one or more defects shall be cause for rejection.

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4.4.1.2 Tests. The preproduction pack of the crated turnout for Level A shall be subjected to one of the following tests. The railroad car impact test shall be used when such facilities are available. The pendulum impact test is an alternate test. At the conclusion of the preproduction pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of holddowns, ties, stays, blocking or bracing, or any visible damage to the contents shall constitute failure of the test and shall be cause for rejection.

4.4.1.2.1 Railroad car impact. Subject the crated turnout to the railroad car test as specified for guided-impact tests in accordance with MIL-STD-1186, Appendix A. The car shall strike a string of five empty cars, with draft gear extended and the brakes set, at a speed of not less than 10 miles per hour (16 kmph).

4.4.1.2.2 Pendulum-impact. Subject the crated turnout to the pendulum-impact test in accordance with FED. TEST METHOD STD. No. 101, Test Method 5012.

4.4.2 Quality conformance inspection of pack.

4.4.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.4.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2.3 Examination. Samples selected in accordance with 4.4.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

105.	Materials or containers not as specified. Each incorrect material or container shall be considered one defect.
106.	Fiberboard box containing the switch lamp not closed, Method V as specified.
107.	Small component parts and accessories not consolidated as specified.
108.	Blocking, bracing, and anchoring not as specified.
109.	Marking illegible, incorrect, incomplete, or missing.

5. PACKAGING

5.1 Preproduction pack. The contractor shall furnish a preproduction pack for examination and testing within the time frame specified (see 6.2) to prove prior to starting production packagingl hat the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination and tests shall be those specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.4).

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5.2 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Unprotected metal surfaces. Unprotected metal surfaces of component parts and accessories of the turnout requiring a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative. The preservative shall conform to the specification listed in, and shall be applied in accordance with MIL-P-116.

5.2.1.2 Switch lamp. The switch lamp shall be removed from the switch stand and shall be preserved in a close-fitting box conforming to PPP-B-636, W6c, style optional. Box closure shall be in accordance with the appendix to the box specification, Method V.

5.2.1.3 Consolidation. Small component parts and accessories such as bolts, nuts, and washers and any other unassembled small components and accessories, including the preserved switch lamp, shall be consolidated in a close-fitting box conforming to PPP-B-601, Domestic Type, style optional. Cushioning and blocking shall be provided to control movement of components and accessories within the box. Strapping of boxes shall not be required.

5.2.2 Commercial. Turnouts shall be preserved in accordance with MIL-STD-1188.

5.3 Packing. Packing shall be Level A or Commercial as specified (see 6.2).

5.3.1 Level A. Component parts, assemblies, and accessories comprising a complete turnout shall be packed in a crate conforming to MIL-C-3774, Type I, Style A. Blocking, bracing, and anchoring of the contents shall be in accordance with the appendix to the crate specification and MIL-STD-1186.

5.3.2 Commercial. Component parts, assemblies, and accessories comprising a complete turnout shall be packed in accordance with MIL-STD-1188.

5.4 Marking.

5.4.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.4.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. Railroad turnouts are intended for diverting rolling stock from one track to another.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size (number) of turnouts required and weight of rail for which it is to be used (see 1.2 and 3.3.1.1).
(c) When closure rails shall be included (see 3.3 and 6.3).
(d) Switch stand required if other than as specified in 3.3.1.
(e) Size of bolts and nuts required (see 3.3.1.2).
(f) Size of joint bars required (see 3.3.1.3).
(g) Size of tie plates required (see 3.3.1.4).
(h) When guard rails are not required (see 3.3.1.5).
(i) Which examination shall be performed by the contractor and which examinations will be performed by the Government if other than as specified in 4.1.
(j) Time frame required for submission of the preproduction pack (see 5.1).
(k) Level of preservation and degree of packing required (see 5.2 and 5.3).

6.3 Closure rails. Closure rails shall not be specified for other than continental United States use where the design calls for a series of turnouts on a yard ladder or similar layout or where economies can be affected in the installation of the turnout in the field (see 3.3 and 6.2).

6.4 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the turnouts in accordance with this specification.

6.5 Recycled material. It is encouraged that recycled materials be used when practical as long as it meets the requirements of the specification (see 3.2).

6.6 Metric measurements. Metric measurements in this specification have been converted and approximated from the inch measurement.

Custodians:
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Project 2250-0053

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

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2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

YES NO (If "Yes", in what way?)

4 REMARKS

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